Code No: 157CE **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
INTRODUCTION TO EMBEDDED SYSTEMS
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

JNTUH Used papers 2023

PART – A

(25 Marks)

1.a)	List some examples of third and fourth generation embedded systems.	[2]
b)	Give some important applications of embedded systems.	[3]
c)	What is the purpose watchdog timer?	[2]
d)	Write the advantages of PLD.	[3]
e)	What is assembly language programming?	[2]
f)	Write the limitations of high level language based development.	[3]
g)	What is the process life cycle?	[2]
h)	What are the merits and demerits of sleep & wakeup mechanism based mutual exclusion?	[3]
i)	What is monitor programming?	[2]
j)	Give the differences between Intel Hex and Motorola Hex file format.	[3]

PART – B

(50 Marks)

2. Explain the various purposes of embedded systems with illustrative examples. [10]

OR

3. What is operational quality attribute? Explain the important operational quality attributes to be considered in any embedded system design. [10]

4. What are the different types of memories used in Embedded System design? Explain the role of each. [10]

OR

5. Explain the role of reset circuit and real time clock in embedded system design. [10]

6. Explain the different Embedded Firmware Design Approaches in detail. [10]

OR

7. What is a pointer in embedded C programming? Explain the different arithmetic and relational operations supported by pointers. [10]

8. Explain the commonly used thread standards for thread creation and management by different operating systems. [10]

OR

9. Explain the various task synchronization techniques. [10]

10. Explain the different tools used for hardware debugging. [10]

OR

11. Explain the Boundary Scan based hardware debugging in detail. [10]

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